

Research article

The effect of education expenditure on air pollution: Evidence from China

Nan Zhao^{a,b}, Chenyang Wang^{a,b,*}, Chunyan Shi^{a,b}, Xiaojie Liu^c^a School of Statistics, Beijing Normal University, No. 19, Xijiekouwai St, Haidian District, Beijing, 100875, PR China^b Center for Education Economics and Statistics of China, No. 19, Xijiekouwai St, Haidian District, Beijing, 100875, PR China^c College of Science, North China University of Technology, Beijing, 100144, PR China

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ABSTRACT

Education expenditure is essential in mitigating air pollution, but the relationship between education expenditure and air pollution lacks in-depth discussion. Utilizing data at the county level in China during 2007–2021, this study estimates the effect of education expenditure from local governments on air pollution. Our findings demonstrate that education expenditure significantly and negatively affects air pollution, which remains robust after addressing endogeneity. The mechanism analysis presents that education expenditure reduces air pollution through the composition, technique, and income effects. The heterogeneity analysis indicates that the impact of education expenditure exhibits marked regional heterogeneity. Specifically, the role of education expenditure is significant in strong regulation, key, eastern, and central regions. By considering interaction terms, we identify the moderating effects of human capital, economic development, infrastructure construction, and public service for education expenditure. The cost-benefit analysis emphasizes that education expenditure improves social welfare. Our findings can inspire local governments to place more emphasis on air quality and public education expenditure.

1. Introduction

Air pollution is a considerable issue in China. Since the reform and opening up, China's economic growth has led to remarkable achievements. However, such rapid economic growth has caused a series of environmental problems, especially air pollution (Zeng et al., 2019). According to China's Ecological and Environmental Status Report published by the Ministry of Ecology and Environment of the People's Republic of China (MEE), only 62.8% of 339 cities in China met the ambient air quality standards in 2022. Air pollution can lead to various detrimental consequences, such as threatening public health (Chen and Chen, 2021), adding medical burdens (Deryugina et al., 2019), and reducing worker productivity (Graff Zivin and Neidell, 2012). Consequently, studying the causes of air pollution has important practical implications.

Fine particulate matter (PM_{2.5}) is a more prominent problem among all kinds of air pollutants. PM_{2.5} refers to fine particulate matter with a diameter of 2.5 μm or less (Yousefi et al., 2023). Excessive PM_{2.5} concentration causes a series of environmental and health challenges (Yi et al., 2019; Zhang et al., 2018). Therefore, PM_{2.5} has attracted widespread attention from the government and the public in China. The

Environmental Air Quality Standard (GB3095-2012) issued by the MEE in 2012 first specified the concentration limits for PM_{2.5}. To reduce PM_{2.5} concentration, the State Council released the Three-year Action Plan to Win Blue Sky Defense War in 2018 (Zhao et al., 2022). Furthermore, air pollution has become a hot research topic, and previous studies mainly focus on the influencing factors and the economic consequences of PM_{2.5} (Wei et al., 2023b). A large body of literature considers PM_{2.5} as the proxy of air pollution (Fu et al., 2021; Wang et al., 2021). As a result, PM_{2.5} concentration is perceived to be air pollution levels in this paper.

There is a rich literature on various drivers of air pollution, such as environmental regulation or energy use (Zeng et al., 2022; Zou, 2021). Besides these determinants, education expenditure also exerts a non-negligible role in mitigating air pollution (Cheng et al., 2021). Specifically, increasing education expenditure by local governments can enhance the level of human capital and strengthen citizens' awareness of environmental protection, thereby reducing energy consumption and air pollution emissions (Mehmood, 2022). Furthermore, raising the proportion of education expenditure in fiscal spending would reduce investment in traditional industries, thus directly alleviating air pollution. As a result, adjusting education expenditure as an effective fiscal tool

* Corresponding author. School of Statistics, Beijing Normal University, No. 19, Xijiekouwai St, Haidian District, Beijing, 100875, PR China.

E-mail address: wcy_econ@163.com (C. Wang).<https://doi.org/10.1016/j.jenvman.2024.121006>

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can achieve the dual benefits of promoting human capital accumulation and improving environmental quality. Compared to other factors, increasing education expenditure can improve air quality through three pathways: composition, technique, and income effects. The composition effect illustrates that increasing education expenditure promotes the transformation from physical capital-intensive industries to human capital-intensive industries (Feng et al., 2022). This industrial structure upgrading resulting from education expenditure alleviates air pollution (Li et al., 2023). In the sight of the technique effect, allocating more government spending on education contributes to the enhancement of human capital levels, thereby promoting the adoption of clean technologies and fostering environmental responsibility awareness (Wei et al., 2023a). Regarding the income effect, more education expenditure implies an improvement in the average income level. As income levels rise, people prioritize environmental quality, leading to a reduction in pollution emissions (Hu and Liao, 2023). Regrettably, there are few systematic analyses of how education expenditure influences air pollution. To address this gap, this paper investigates the effect of education expenditure by county governments on air pollution in China. Our findings can provide policy enlightenment for governments in emphasizing education expenditure and improving air quality.

Our study contributes to the literature in three ways. First, instead of focusing on education achievements in most of the previous literature (Balaguer and Cantavella, 2018; Zafar et al., 2020), we shed new light on the impact of education expenditure by local governments on air pollution. Although some studies involve the role of education expenditure, they do not conduct an in-depth analysis. To the best of our knowledge, only Hua et al. (2018) conducted a prefecture-level analysis of the relationship between air pollution and education expenditure in China. We attempt to target this deficiency and fill the gap through empirical analysis.

Second, we provide causal evidence for investigating the effect of education expenditure on air pollution. The existing literature places little emphasis on the endogeneity problem, which is inevitable between education expenditure and air pollution. We address the endogeneity problem based on the instrumental variable method. Additionally, some studies only discuss the impact of education expenditure (Bulus and Koc, 2021; Fernández et al., 2018), and we consider the pathway through which education expenditure affects air pollution.

Third, for the first time, we explore the effect of education expenditure on air pollution from a county-level perspective. Due to data limitations, previous studies generally exploit data at the national or prefectural level (Lin and Zhu, 2019; Zaman et al., 2021). We utilize nationwide county-level panel data and identify heterogeneity caused by regulation intensity, the severity of air pollution, and geographical location. Moreover, this paper examines the moderating effects of several socioeconomic factors at the county level on education expenditure.

The remainder of this paper is organized as follows. Section 2 contains the literature review and research hypotheses. Section 3 establishes the model and outlines the data sources. Section 4 shows the results. Section 5 presents the mechanism analysis. Section 6 conducts further discussion. Section 7 summarizes the conclusions and policy implications.

2. Literature review and research hypotheses

2.1. The effect of education expenditure on air pollution

Many socioeconomic factors are recognized as crucial determinants of air pollution (Lamla, 2009). The most trailblazing theory is the Environmental Kuznets Curve (EKC) hypothesis, which postulates an inverted U-shaped relationship between pollutants and per capita income (Grossman and Krueger, 1995). Subsequently, substantial literature has provided empirical evidence on how socioeconomic factors such as education (Bano et al., 2018), international trade (Levinson,

2009), and technological innovation (Chen et al., 2022a) influence air quality. Education factors, including educational input and achievement, occupy a dominant position among all socioeconomic factors of air pollution (Li and Ouyang, 2019). Previous studies have focused on the role of educational achievement and concluded that upgrading the human capital level can improve air quality (Sarkodie et al., 2020; Yao et al., 2020). Regrettably, the relationship between education expenditure and air pollution still lacks systematic and comprehensive discussion.

Air pollution is a public good with negative externalities, and market failure can hardly be avoided because of the externalities. Therefore, improving air quality requires government intervention (Jiang et al., 2021). Education expenditure, accounting for a large proportion of total fiscal spending, serves as one of the main tools of government intervention. Local governments investing more in education can simultaneously obtain educational returns and combat environmental problems. Therefore, a growing body of studies has begun to focus on the role of education expenditure on air pollution but has not reached a consensus (Liu et al., 2023; Tang et al., 2023). Zaman et al. (2021) demonstrated that education expenditure directly reduces pollution emissions, thus the government should guarantee budgets for education to raise public environmental awareness. Conversely, Xue et al. (2023) found that ensuring education expenditure would amplify the financial pressure on the government, thereby enhancing carbon intensity. In addition, Ozyilmaz et al. (2023) argued that there is no significant relationship between education expenditure and environmental pollution through empirical cross-country data analysis. Based on the above analysis, in order to test whether education expenditure has an impact on air pollution, we formulate the first hypothesis of this paper:

Hypothesis 1. (H1): Education expenditure has a negative effect on air pollution.

2.2. Three mechanisms by which education expenditure affects air pollution

2.2.1. Composition effect

The composition effect reveals that education expenditure alleviates air pollution by adjusting industrial structure (López et al., 2011; Zhang et al., 2021). On the one hand, increasing education expenditure will bring about a shift in the industrial structure. Education expenditure tends to exert a role in human capital-intensive as opposed to physical capital-intensive activities (Tong et al., 2020). Accordingly, raising education expenditure will lead to a diminished proportion of the secondary industry in GDP and a higher proportion of the tertiary industry (Yumei et al., 2022). On the other hand, compared to the primary and tertiary industries, the air pollution issue is most severe within the secondary industry in China (Muhammad et al., 2022). The secondary industry, comprising mining, manufacturing, and electricity, emits a volume of pollutants during the production process (Guan et al., 2016). Upgrading the industrial structure would reduce air pollutant emissions and improve environmental quality (Wang et al., 2023). Therefore, augmenting education expenditure can alleviate air pollution by changing the industrial structure. In light of the above analysis, we propose the following hypothesis:

Hypothesis 2a. (H2a): Education expenditure affects air pollution through the composition effect.

2.2.2. Technique effect

The technique effect illustrates that education expenditure improves air quality by raising the level of human capital. Increasing fiscal spending on public education facilitates the cultivation of more talent (Samuels and Vargas, 2023), and the enhancement of human capital primarily contributes to reducing air pollution in two ways. First, highly skilled people are key drivers of accelerating technological progress (Ali et al., 2021). Technological progress implies cleaner production patterns

and increased energy efficiency (Voigt et al., 2014). The shift in production patterns directly mitigates pollutant emissions and alleviates air pollution. Second, higher levels of education correspond to a stronger sense of social responsibility. Well-educated individuals will be more environmentally conscious in their production and living processes, and they are likely to make economic decisions with greater consideration of environmental consequences (Sun et al., 2021). Based on the above analysis, we propose the following hypotheses:

Hypothesis 2b. (H2b): Education expenditure affects air pollution through the technique effect.

2.2.3. Income effect

The income effect indicates that raising public education investment affects air pollution by influencing average national income levels (Zhang et al., 2017). First, allocating more fiscal spending toward education results in elevating the overall education level (Johnson and Jackson, 2019). Enhancing overall educational attainment leads to prosperous economic growth and higher average income levels. Furthermore, as income levels rise, individuals are inclined to pursue a high-quality living environment, particularly in terms of ecological surroundings. To be specific, people tend to adopt an eco-friendly lifestyle to promote environmental quality (Mehmood, 2021). This trend would bring about resource conservation and the application of green technologies, which contribute to reducing pollutant emissions. Consequently, higher individuals' income levels mean less air pollution emission. According to the above discussion, the increase in income levels resulting from more education investment might decrease air pollution emissions. Hence, we propose the following hypotheses:

Hypothesis 2c. (H2c): Education expenditure affects air pollution through the income effect.

3. Methodology and data

3.1. Model specification

This paper aims to estimate the effect of education expenditure on air pollution. Based on the above theoretical analysis, we establish a two-way fixed effects model as the baseline regression:

$$Mean_{ct} = \beta_0 + \beta_1 Edu_{ct} + \beta_2 Control1_{ct} + \beta_3 Control2_{pt} + \mu_c + \lambda_t + \varepsilon_{ct} \quad (1)$$

where the subscript c represents the county, and t represents the year. $Mean$ is the dependent variable representing the annual mean of $PM_{2.5}$ concentration. Edu is the key independent variable standing for education expenditure from county governments. $Control1$ and $Control2$ are vectors of control variables at the county and city levels, respectively. To prevent omitting factors that are unobservable and do not vary over time, we control for county fixed effects μ_c . λ_t is the year fixed effect, which captures the time trend. ε_{ct} denotes the error term.

3.2. Description of variables

3.2.1. Dependent variable

Based on the preceding analysis, we consider $PM_{2.5}$ concentration as the proxy of air pollution levels. Specifically, the dependent variable is the arithmetic average of the daily mean of $PM_{2.5}$ concentration for 365 days in a year.

3.2.2. Key independent variable

The key independent variable in the paper is basic education expenditure from county governments. Basic education consists of preschool, primary school, middle school, and high school education in China (Liu et al., 2020). Remarkably, in China's education finance system, county governments are responsible for basic education finance, and most of the education expenditure from county governments is

allocated to basic education (Jia et al., 2014). Therefore, considering the fact that China's basic education finances are dominated by county governments, we choose basic education expenditure as the key independent variable.

3.2.3. Control variables

With reference to prior studies, we introduce the following county-level and city-level control variables that may affect air pollution. Three county-level control variables are included: **Economic development.** In light of the EKC hypothesis, economic development is a crucial determinant of air pollution (Ouyang et al., 2019). Following He (2015), we include the regional gross domestic product (GDP) per capita ($pGDP$) and its squared term ($pGDP^2$) in control variables. **Fiscal spending.** Fiscal instrument is a powerful approach for mitigating environmental degradation (Ullah et al., 2020). Based on Halkos and Paizanos (2013), we control the total fiscal spending ($Fiscal$). **Population density.** The population density of a region has a positive impact on the $PM_{2.5}$ concentration (Ouyang et al., 2021). We adopt population density ($Popden$), measured as the proportion of the population to the area of each county, to control for the effect of population density (Chen et al., 2020). Two city-level control variables are included: **Scale of transportation.** The transportation is a major source of air pollution emissions (Jia et al., 2021), thus we employ highway passenger traffic ($Traffic$) as the proxy for the scale of transportation (Chen et al., 2022b). **Energy use.** Improving energy efficiency can effectively alleviate air pollution (Miao et al., 2019). To quantify the energy use in the city, we consider the proportion of energy consumption to GDP as the control variable (Wen and Liu, 2022), denoted as energy intensity ($Energy$). However, due to the official energy consumption data in the city is unavailable (Wang and Wang, 2020), we thus obtain the city-level energy consumption by calculating the product of provincial energy consumption and the ratio of the city-level nighttime lighting values in the province (Li et al., 2021).

3.3. Data sources and descriptive statistics

We collect and combine three data sources. First, $PM_{2.5}$ concentration data from 2007 to 2021 is obtained from the Atmospheric Composition Analysis Group at Dalhousie University. We use annual mean $PM_{2.5}$ concentration data for each county as the explained variable, which is in units of micrograms per cubic meter ($\mu g/m^3$).

Second, the education expenditure data from county governments is derived from the Center for Education Economics and Statistics of China. We have detailed data on education information from preschool to higher education at the county level in China, including education expenditure, number of students and teachers, etc. Notably, compared with previous years, the fiscal revenue and expenditure accounts experienced a major reform in China in 2007. In particular, the statistical calibers of fiscal spending items changed considerably. Given the comparability of data, we only collect education expenditure data for 2007 and beyond.

Third, control variables at the county and city levels come from the yearbook data released by the National Bureau of Statistics (NBS). Provincial energy consumption data is sourced from the China Energy Statistical Yearbook. The nighttime data is from the National Geophysical Data Center of the National Oceanic and Atmospheric Administration in the US.

All funding data is deflated using the GDP index at the provincial level, taking 2007 as the base period. Furthermore, all continuous variables are winsorized at the 2% and 98% levels. Eventually, we obtain unique and unbalanced panel data from 1706 counties across 28 provinces over the 2007–2021 period. The descriptive statistics of the main variables are presented in Table 1.

Table 1
Descriptive statistics of main variables.

Variable	Definition	N	Mean	Std.	Min	Max
<i>Mean</i>	Annual mean of PM _{2.5} concentration ($\mu\text{g}/\text{m}^3$)	21,515	44.835	17.827	11.474	128.226
<i>Edu</i>	Education expenditure (100 million yuan)	21,515	2.226	1.627	0.195	10.299
<i>Fiscal</i>	Total fiscal spending (100 million yuan)	21,515	11.582	7.941	0.933	78.573
<i>pGDP</i>	Per capita GDP (10,000 yuan/per person)	21,515	1.692	1.462	0.325	7.786
<i>Popden</i>	Population density (10,000 persons/ km^2)	21,515	0.516	0.870	0.004	7.196
<i>Traffic</i>	Ln (Highway passenger traffic+1) (100 million persons)	21,515	0.506	0.431	0.001	3.390
<i>Energy</i>	Proportion of energy consumption to GDP	21,515	1.043	0.856	0.135	12.808
<i>Ind2str</i>	The share of second industry in GDP	21,515	0.431	0.152	0.000	0.746
<i>Ind3str</i>	The share of tertiary industry in GDP	21,515	0.369	0.122	0.000	0.702
<i>HTea</i>	Number of senior high school teachers (1000 persons)	21,515	0.905	0.659	0.000	6.626
<i>Internet</i>	Subscribers of Internet services per million persons	18,359	16.292	13.282	2.092	69.652
<i>Tele</i>	Number of fixed-line telephone users (1000)	20,352	65.859	61.298	2.492	254.865
<i>pBeds</i>	Number of hospital beds per thousand persons	21,515	3.667	1.912	0.873	10.111
<i>Welfare</i>	Number of welfare institutions	20,388	19.470	21.213	1.000	121.000

Note: All expenditure variables are adjusted by the provincial GDP index.

4. Results

4.1. Baseline regression results

We first examine the effect of education expenditure on air pollution. The regression results of Eq. (1) are shown in Table 2. Given that the PM_{2.5} concentration in the previous year might have a positive effect on the PM_{2.5} concentration in the current year, we add a lagged term of the dependent variable (*L.Mean*) as a control variable to capture the dynamic changes in air pollution. Table 2 displays baseline regression results. Columns (1) and (3) of Table 2 report the regression results without control variables, and columns (2) and (4) report the results with control variables. From all columns of Table 2, we find that the coefficients of education expenditure are always negative and significant at the 1% level. The baseline regression results demonstrate that education expenditure has a negative and significant effect on the annual mean concentration of PM_{2.5}. In addition, the adjusted R-squared values of all results are greater than 0.9, indicating that the models we

Table 2
Baseline regression results.

Variables	(1)	(2)	(3)	(4)
	<i>Mean</i>	<i>Mean</i>	<i>Mean</i>	<i>Mean</i>
<i>Edu</i>	−1.193*** (0.145)	−0.942*** (0.145)	−0.556*** (0.089)	−0.400*** (0.090)
<i>L.Mean</i>			0.596*** (0.011)	0.583*** (0.011)
<i>Fiscal</i>		−0.040* (0.023)		−0.033** (0.015)
<i>pGDP</i>		1.188*** (0.420)		0.381 (0.241)
<i>pGDP2</i>		−0.156*** (0.047)		−0.060** (0.028)
<i>Popden</i>		−7.756*** (1.573)		−4.845*** (1.067)
<i>Traffic</i>		3.299*** (0.356)		1.521*** (0.215)
<i>Energy</i>		0.314*** (0.054)		0.148*** (0.042)
Constant	47.490*** (0.323)	48.171*** (1.074)	18.311*** (0.506)	20.132*** (0.846)
County effect	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes
Observations	21,515	21,515	19,379	19,379
Adjusted R-squared	0.927	0.928	0.949	0.949

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

construct fit well. Our findings preliminarily confirm **H1** in the literature review and are consistent with previous findings at the prefectural level (Hua et al., 2018).

4.2. Endogeneity

Although the baseline regression results are in line with our expectations, possible endogeneity problems are unavoidable in Eq. (1). Endogenous challenges may arise for three reasons. (1) Reverse causality. Guo et al. (2024) indicate that air pollution has a significant and negative impact on education expenditure. Therefore, there is a reverse causal relationship between air pollution and education expenditure. (2) Omitted variables. The data at the county level from 2007 to 2021 are difficult to acquire, and we may omit some control variables that might affect air pollution. Among the factors we omit, those correlated with education expenditure would influence consistent estimation for our models. (3) Measurement error. The dependent variable in this paper is measured data, and there may be measurement error issues. Limited to our literature search results, the endogeneity problem of education expenditure as the key independent variable has not received sufficient attention. Some studies use lagged terms as instrumental variables but may not obtain consistent estimation coefficients (Bellemare et al., 2017). To fill this gap, we employ three more appropriate instrumental variables and estimate two-stage least squares (2SLS) regression models separately.

First, following Jackson et al. (2016), we exploit a major reform of education finance in China as an instrumental variable. Specifically, to ensure the scale of fiscal spending on education, the Ministry of Education formulated the Outline of China's National Plan for Medium- and Long-term Education Reform and Development (2010–2020) ("Outline") in 2010.¹ The "Outline" explicitly stated that "the proportion of fiscal spending on education in GDP should not be less than 4% by 2012" ("4% target"). This reform satisfies both the requirement of relevance and exogeneity of instrumental variables. In terms of

¹ Data source: http://www.moe.gov.cn/srcsite/A01/s7048/201007/t20100729_171904.html?gs_ws=tqq_635879677144434007.

relevance, the promulgation of the “Outline” is a momentous reform in the history of China’s education finance development during 2007–2021 (Han and Ye, 2017). Governments at all levels in China have attached great priority to the “4% target” and increased fiscal spending on education to achieve this target. Consequently, the implementation of this reform is highly and positively correlated with education expenditure from county governments. Regarding exogeneity, this policy is not aimed at mitigating air pollution and can be considered entirely exogenous. The first stage is a difference-in-differences (DID) model with the following expression:

$$Edu_{ct} = \alpha_0 + \alpha_1 Treat_{c,2010} \times Time_t + \alpha_2 Control1_{ct} + \alpha_3 Control2_{pt} + \mu_c + \lambda_t + \xi_{ct} \quad (2)$$

where *Treat* stands for the intensity of the policy. *Time* is a dummy variable, taking the value of 1 for 2011 and later, and 0 otherwise. ξ_{ct} is the error term. Other variables have the same meanings as in Eq. (1). As for the measurement for *Treat*, we calculate the proportion of education expenditure in GDP in 2010 for each county, denoted as *EduRate*. According to the “4% target”, if the value of *EduRate* is greater than or equal to 0.04, we consider the intensity of the policy to be 0. If the value of *EduRate* is less than 0.04, the difference between it and 0.04 is defined as the policy intensity. The specific expression of *Treat* is expressed in Eq. (3).

$$Treat_{c,2010} = \begin{cases} 0.04 - EduRate_{c,2010}, & EduRate_{c,2010} < 0.04 \\ 0, & EduRate_{c,2010} \geq 0.04 \end{cases} \quad (3)$$

The estimated results of the 2SLS-DID model are shown in columns (1) and (2) of Table 3. The F-statistic of the first stage is larger than 10, which indicates that the first instrumental variable passes the weak instrumental variable test. The results in column (1) of Table 3 show that education expenditure sharply increases after the reform is implemented. The results in column (2) of Table 3 present that education expenditure still has a significant and negative effect on PM_{2.5} concentration after addressing the endogeneity problem.

The parallel trend test is an essential prerequisite for using a DID model. Following Liu and Mao (2019), we establish the following model for parallel trend test.

$$Edu_{ct} = \gamma_0 + \gamma_t \sum_{t \neq 2010} Treat_{c,2010} \times Time1_t + \gamma_2 Control1_{ct} + \gamma_3 Control2_{pt} + \mu_c + \lambda_t + \epsilon_{ct} \quad (4)$$

where *Time1_t* is a year dummy variable, taking the value of 1 when the year is *t*, and 0 otherwise. ϵ_{ct} is the error term. Other variables have the same meaning as in Eq. (1). Fig. 1 displays the results of the parallel

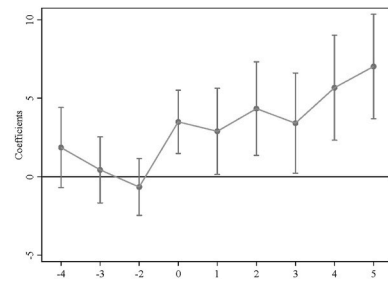


Fig. 1. Parallel trend test.

trend test. Taking 2010 as the base period, the coefficients of previous years are not significant. After the implementation of the policy, the coefficients of each item are significantly not 0, which means that the parallel trend test has been passed. In addition, we conduct a placebo test to eliminate the influence of other policies. Specifically, we randomly generate 500 treatment groups and repeat the regressions 500 times. We then obtain the kernel density change of the t value corresponding to 500 regressions and compare them with the t value of the education expenditure in column (2) of Table 3. Fig. 2 reveals that none of the placebo test in the randomly generated 500 treatment groups produce more significant estimates than the result in column (2) of Table 3, which suggests that our DID results pass the placebo test.

The second instrumental variable is the number of basic education schools in each county (*InsNum*). The vast majority of education expenditure by county governments is spent on basic education, thus there should be a positive correlation between the number of schools and the total amount of education expenditure. Moreover, we have already controlled for *Popden* in the model, and other channels through which *InsNum* could affect PM_{2.5} concentration are severed. Therefore, *InsNum* can satisfy the exogeneity of instrumental variables. The

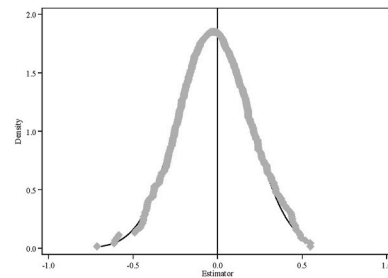


Fig. 2. Placebo test.

Table 3
Results of the instrumental variable approach.

Variables	(1) <i>Edu</i>	(2) <i>Mean</i>	(3) <i>Edu</i>	(4) <i>Mean</i>	(5) <i>Edu</i>	(6) <i>Mean</i>
<i>Edu</i>		−7.610*** (2.148)		−6.864*** (1.113)		−5.611*** (0.576)
<i>Treat</i> × <i>Time</i>	5.793*** (1.396)					
<i>InsNum</i>			0.001*** (0.000)			
<i>IV</i>					0.654*** (0.039)	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap rk Wald F statistic	17.223		80.206		282.307	
Observations	21,515	21,515	21,515	21,515	21,515	21,515

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

Table 4

The results for robustness tests.

Variables	(1) Mean	(2) Mean	(3) Mean	(4) Mean	(5) lMean
<i>Edu</i>	−0.947*** (0.149)				
<i>pEdu</i>		−4.122*** (0.919)			
<i>L.Edu</i>			−0.446*** (0.129)		
<i>L2.Edu</i>				−0.544*** (0.155)	
<i>lEdu</i>					−0.075*** (0.009)
Control variables	Yes	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes	Yes
Observations	20,810	21,515	19,379	17,721	21,515
Adjusted R-squared	0.928	0.852	0.927	0.924	0.954

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

regression results are presented in columns (3) and (4) of Table 3, and the conclusion of H1 holds firm.

Third, we attempt to exploit the mean education expenditure of each county in a province except for itself (*IV*) as an instrumental variable. Columns (5) and (6) of Table 3 report the results, which indicate that education expenditure negatively affects air pollution after addressing the endogeneity problem.

4.3. Robustness test

We use three approaches to ensure the robustness of our conclusions. First, given that the management systems of municipalities are quite different from other provinces, we remove the sample belonging to four municipalities: Beijing, Tianjin, Shanghai, and Chongqing. The results are displayed in column (1) of Table 4. As we can see, the estimated coefficient of the key independent variable remains negative and significant at the 1% level.

Second, we substitute the key independent variable to test the robustness. The per pupil education expenditure is also an appropriate indicator reflecting the effort on education by the county-level government (Huang et al., 2017). We adopt the per pupil education expenditure (*pEdu*) as the key independent variable and conduct the regression again. The regression results in column (2) of Table 4 indicate that the estimated coefficient of per pupil education expenditure is significantly negative, which supports our main conclusions. Furthermore, taking into account the potential reverse causation and the lag effect of education expenditure, we lag the key independent variable to conduct the robustness test. Following Wang and Tang (2023), we use the key independent variables lagged by one or two periods (*L.Edu* and *L2.Edu*) as substitutes for the key independent variable of the current period. As presented in columns (3) and (4) of Table 4, the lag terms of education expenditure have a negative and significant effect on PM_{2.5} concentration.

Third, we employ a double logarithmic model for the robustness test. We logarithmize both the dependent variable (*lMean*) and the key independent variable (*lEdu*). The other variables are left unchanged, and the regression is performed again. The regression results are presented in column (5) of Table 4. The estimation results remain supportive of our main conclusions.

5. Mechanism analysis

The above results confirm H1 and indicate that education expenditure has a negative effect on air pollution. Another focus of this paper is the mechanism behind the role of education expenditure in air pollution.

To validate H2a–H2c, we attempt to replace the dependent variable in Eq. (1) as the mechanism variable sequentially to verify the channels through which education expenditure may affect air pollution.

5.1. Composition effect

In order to verify H2a, we study how education expenditure influences the industrial structure. We take the share of the secondary industry on GDP (*Ind2str*) and the share of the tertiary industry on GDP (*Ind3str*) as the explained variables to identify the composition effect, respectively. The results are shown in columns (1) and (2) of Table 5. The results demonstrate that education expenditure can significantly reduce the proportion of the secondary industry while increasing the share of the tertiary sector. Therefore, H2a in this paper has been validated. Our findings provide implications for developing countries in achieving environmental protection through structural transformation. Economies in developing countries are undergoing a profound structural transformation, and the advancement of the tertiary industry is crucial to improving environmental quality (Yu and Wang, 2021).

5.2. Technique effect

To test H2b, we explore the effect of education expenditure on human capital levels. We consider the number of senior high school students in each county (*HStu*) as the proxy of human capital. Column (3) of Table 5 reports the regression results. The estimated coefficient of *Edu* is positive and significant at the 1% level, which reveals that education expenditure reduces air pollution through the technique effect and confirms H2b. As public education investment intensifies, the rise in human capital levels leads to an influx of highly skilled individuals into the labor market (Li et al., 2019; Quan et al., 2021). Our empirical results emphasize the environmental benefits of human capital.

5.3. Income effect

To test H2c, we replace the explained variable of Eq. (1) with resident income. We use the balance of household savings deposits (*Deposit*) as the proxy variable of resident income. The estimated results presented in column (4) of Table 5 demonstrate that education expenditure has a positive effect on resident income, which supports H2c. Our results validate that income motivates individuals to have higher expectations for environmental quality and to adopt environmentally-friendly lifestyles.

Table 5
The results for mechanism analysis.

Variables	(1)	(2)	(3)	(4)
	<i>Ind2str</i>	<i>Ind3str</i>	<i>HStu</i>	<i>Deposit</i>
<i>Edu</i>	−0.004** (0.002)	0.004** (0.002)	0.794*** (0.156)	4.097*** (0.453)
Control variables	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes
Observations	21,515	21,515	21,515	20,092
Adjusted R-squared	0.814	0.712	0.916	0.968

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

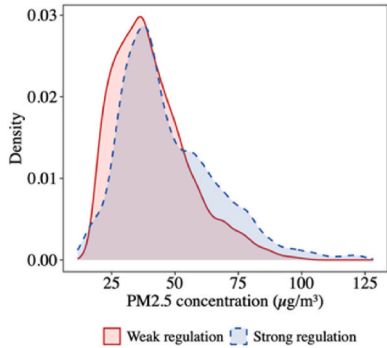


Fig. 3. Kernel density curves of weak and strong regulation regions.

6. Further discussion

The further discussion includes two parts, heterogeneity analysis and moderating effect analysis. Does the impact of education expenditure on air pollution vary according to factors such as pollution severity or geographical distribution of the area? Do economic development, human capital, and other socioeconomic factors play moderating effects on the impact of education expenditure? This section attempts to answer these questions through empirical analyses.

6.1. Heterogeneity analysis

First, environmental regulation is a crucial determinant of air pollution (Shapiro and Walker, 2018), and regional variations in regulatory intensity lead to differential impacts on education expenditure. To investigate this, we conduct two subsample regressions based on the intensity of environmental regulation, which is measured by the proportion of investment in industrial pollution control to industrial added value (Song et al., 2021). Further, according to the upper and lower tertiles of this proportion, the full sample is divided into “strong regulation” and “weak regulation” regions, and the corresponding kernel density curves are displayed in Fig. 3. As expected, PM_{2.5} concentration in weak regulation regions is lower than strong regulation regions. In Table 6, we report the results of the two subsample regressions in columns (1) and (2). These results indicate that education expenditure significantly and negatively affects PM_{2.5} concentration only in regions with strong regulation, whereas the coefficient is not statistically significant in weak regulation regions. From these findings, we may infer that implementing environmental regulations contributes to the role of education expenditure in mitigating air pollution. This insight suggests that the government should not only enforce environmental regulations robustly but also ensure adequate education expenditure to effectively reduce PM_{2.5} concentration.

Second, the MEE has noted that air pollution in China is not a problem of one region individually but manifests regional characteristics.

Table 6
The results for heterogeneity analysis.

Variables	(1)	(2)	(3)	(4)
	Weak regulation	Strong regulation	Key regions	Non-key regions
	<i>Mean</i>	<i>Mean</i>	<i>Mean</i>	<i>Mean</i>
<i>Edu</i>	0.201 (0.188)	−0.805*** (0.221)	−1.041*** (0.219)	−0.611*** (0.143)
Control variables	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes
Observations	6962	6799	6918	14,597
Adjusted R-squared	0.928	0.946	0.931	0.918
Fisher's Permutation P-value			0.000***	

Note: The two groups of subsamples are divided according to the regulatory intensity and the key regions determined by the MEE. Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

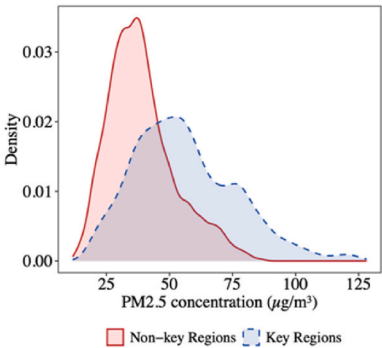


Fig. 4. Kernel density curves of key regions and non-key regions.

Accordingly, the Report on Ecological and Environmental Conditions (2021) has designated three key regions, namely, Jing-Jin-Ji and surrounding areas, the Yangtze River Delta region, and the Fenwei Plain. The specific cities and provinces are listed in Table A1 of Appendix A. The counties belonging to these three key areas are labeled “key regions”, and the remaining are labeled “non-key regions”. According to this criterion, we divide the whole sample into key regions and non-key regions. The kernel density curves of the two subsamples are displayed in Fig. 4, which indicates that PM_{2.5} concentrations in key regions obviously exceed those in non-key regions. The results of the two subsamples are shown in columns (3) and (4) of Table 6, which demonstrate that the estimated coefficients of education expenditure are both negative and significant at the 1% level. To this end, we further test the difference of the estimated coefficients between key and non-key regions using Fisher's Permutation test (Lu et al., 2023; Zhou et al., 2023). The results in the last row of Table 6 present that the empirical P-value is 0.000 with bootstrapped repetitions of 1000, which means that education expenditure exerts a stronger negative role on air pollution in key regions. This heterogeneous result suggests that in the key regions with severe air pollution problems, both the government and the public prioritize environmental quality through various measures, such as environmental regulation and information disclosure. Therefore, education expenditure exerts a more significant role in mitigating air pollution.

Finally, we focus on whether our findings have geographical heterogeneity. Based on the criteria of NBS,² we divide the 31 provinces

² Data source: http://www.stats.gov.cn/zt_18555/zthd/sjtjr/dejtkjfr/tjzp/202302/t20230216_1909741.htm.

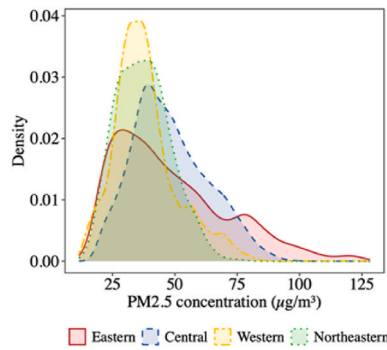


Fig. 5. Kernel density curves of eastern, central, western, and north-eastern regions.

into four regions: eastern, central, western, and northeastern regions. The eastern region includes Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan; the central region encompasses Shanxi, Anhui, Jiangxi, Henan, Hubei, and Hunan; the western region covers Neimenggu, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Tibet, Shaanxi, Gansu, Qinghai, Ningxia and Xinjiang; and the northeastern region includes Liaoning, Jilin and Heilongjiang. The kernel density curves of the four subsamples are shown in Fig. 5. The regression results of the four subsamples are reported in Table 7. The coefficients of the key explanatory variables are still significant in the eastern and central regions, but are no longer significant in the western and northeastern regions. The kernel density curves of Fig. 5 indicate that there is little difference in the distribution of air pollution among the four regions. Thus, we speculate that this variation in significance among the four regions could be interpreted from a socioeconomic perspective. Compared with the western and northeastern regions, the eastern and central regions are more economically advanced, with higher levels of income and human capital (Zhang and Zhuang, 2011). We believe that preexisting advantages may make education input from local governments more efficient.

6.2. Moderating effect

Besides education expenditure, air pollution can be influenced by various socioeconomic factors. Do these factors have effects on the role of education expenditure? To answer this question, we conduct a moderating effect analysis. Specifically, we select four moderating variables: human capital, economic development, infrastructure construction, and public services. We then add interaction terms between education expenditure and these moderating variables into the regression model (Frankel and Rose, 2005). The significance and sign of the estimated coefficients of the interaction terms are our focus.

First, the number of high school teachers can serve as a proxy for the level of human capital (Nguyen and Canh, 2020). Thus, we add the number of senior high school teachers at the county level (*HTea*) and its interaction with education expenditure into Eq. (1). As seen in column (1) of Table 8, the coefficient of the interaction term between *Edu* and *HTea* is significant and negative. The results indicate that a higher level of human capital enhances the negative effect of education expenditure on air pollution.

Second, we analyze the moderating effect of the economic development level. We remove *pGDP2* from control variables in Eq. (1) and add the interaction term of education expenditure and per capita GDP. The results in column (2) of Table 8 show that the coefficient of the interaction term is significantly positive at the 10% level. Specifically, economic development has two opposing effects on air pollution: (1) economic growth leads to more severe pollution; (2) the adequate finances raised from economic growth can alleviate air pollution. In our study, the impact of economic growth on increasing air pollution

Table 7

The results for heterogeneity analysis.

Variables	(1)	(2)	(3)	(4)
	Eastern	Central	Western	Northeastern
	Mean	Mean	Mean	Mean
<i>Edu</i>	−0.779*** (0.209)	−0.780* (0.466)	−1.143 (0.660)	0.230 (0.261)
Control variables	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes
Observations	6734	6166	6303	2312
Adjusted R-squared	0.941	0.827	0.918	0.922

Note: The division of this subsample is based on the regional division of China as defined by the NBS. Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

dominates our results. Accordingly, economic development weakens the positive effect of education expenditure in mitigating $PM_{2.5}$.

In what follows, we explore the moderating effect of the construction of basic telecommunications facilities on the role of education expenditure. Initially, we utilize the number of network users, that is the number of per capita subscribers of Internet services (*Internet*) as a proxy variable (Habibi and Zabardast, 2020). However, the data of *Internet* is only available at the city level for 2007–2019. Consequently, we also consider the number of fixed-line telephone users at the county level (*Tele*) as another proxy variable for the construction of basic telecommunications facilities (Qin et al., 2023). The regression results for these two proxy variables *Internet* and *Tele* are summarized in columns (3) and (4) of Table 8, respectively. We find that the coefficients of the interaction terms are both significantly negative, indicating that the development of infrastructure construction contributes to the greater role of education expenditure.

Finally, we examine the moderating effect of the public service level in each county. The number of hospital beds per capita can capture the level of development of a county's healthcare system and the ability to respond to health emergencies (Porto et al., 2022). Additionally, the number of welfare institutions reflects the social welfare level of the county. We use the number of hospital beds per thousand inhabitants (*pBeds*) and the number of welfare institutions (*Welfare*) as proxy variables of public service level. Columns (5) and (6) of Table 8 display the regression results, and the interaction terms are both significantly negative. Results in columns (5) and (6) of Table 8 demonstrate that the improvement in the public service level has a positive moderating effect on the role of education expenditure.

6.3. Cost-benefit analysis

The cost-benefit analysis provides a quantitative basis for optimizing policy design via monetizing costs and benefits (Zhu et al., 2022). In what follows, we perform a cost-benefit analysis of education expenditure.

Regarding the benefit, mitigating air pollution brings various economic benefits, such as improvements in public health and enhanced labor productivity (Giannadaki et al., 2018). In our paper, the economic benefit considered is the monetary value of the total benefit resulting from the extended life expectancy due to the reduction in $PM_{2.5}$ concentrations. To be more specific, we substitute the dependent variable in Eq. (1) with the logarithmic form of the annual average $PM_{2.5}$ concentration (*lMean*), and the corresponding regression equation is:

$$lMean_{ct} = \theta_0 + \theta_1 Edu_{ct} + \theta_2 Control1_{ct} + \theta_3 Control2_{pt} + \mu_c + \lambda_t + \zeta_{ct} \quad (5)$$

where the definitions of the variables in Eq. (5) are similar to Eq. (1).

The results in column (1) of Table 9 indicate that the estimated coefficient of education expenditure is significant at the 1% level and its

Table 8

The results for moderating effects for socioeconomic variables.

Variables	(1) Mean	(2) Mean	(3) Mean	(4) Mean	(5) Mean	(6) Mean
<i>Edu</i>	−1.006*** (0.311)	−1.821*** (0.236)	−0.511*** (0.131)	−1.352*** (0.201)	−1.105*** (0.212)	−0.786*** (0.163)
<i>Edu</i> × <i>HTea</i>	−0.439** (0.200)					
<i>HTea</i>	0.060 (0.796)					
<i>Edu</i> × <i>pGDP</i>		0.126* (0.064)				
<i>pGDP</i>		−0.336 (0.217)				
<i>Edu</i> × <i>Internet</i>			−0.006*** (0.002)			
<i>Internet</i>			0.025*** (0.009)			
<i>Edu</i> × <i>Tele</i>				−0.001* (0.001)		
<i>Tele</i>				0.007** (0.003)		
<i>Edu</i> × <i>pBeds</i>					−0.067** (0.034)	
<i>pBeds</i>					−0.534*** (0.099)	
<i>Edu</i> × <i>Welfare</i>						−0.005*** (0.002)
<i>Welfare</i>						0.055*** (0.005)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
County effect	Yes	Yes	Yes	Yes	Yes	Yes
Year effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,515	21,515	18,346	20,340	21,515	20,384
Adjusted R-squared	0.929	0.927	0.940	0.928	0.930	0.930

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index. Due to the lack of data, the regression results in column (3) are based on municipal data from 2007 to 2019.

Table 9

Results of the semi-logarithmic model.

Variables	(1) lMean
<i>Edu</i>	−0.0178*** (0.002)
Control variables	Yes
County effect	Yes
Year effect	Yes
Observations	21,515
Adjusted R-squared	0.954

Note: Robust standard errors clustered at the county level appear in parentheses. Significance levels *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All expenditure variables are adjusted by the provincial GDP index.

value is −0.0178, which implies that one unit (100 million yuan) rise in education expenditure is associated with a 1.78% decline in $PM_{2.5}$ concentration. Therefore, education expenditure reduces $PM_{2.5}$ concentration by 0.798 (= $44.835 \times 1.78\%$) $\mu g/m^3$. Furthermore, according to the latest Air Quality Life Index (AQLI) in 2017, life expectancy is extended by 0.098 years for each unit ($\mu g/m^3$) decrease in $PM_{2.5}$ concentration.³ Thus, the reduction of $PM_{2.5}$ concentration increases life expectancy by 0.078 (= 0.798×0.098) years. Based on the data from the NBS, the total labor force was 11.784 billion people from 2007 to 2021. To ensure the benefit is estimated at the lower bound (Wu et al., 2022), we consider the national average minimum wage of 18.1 thousand yuan during the sample period. Finally, the total benefit is 16,637 (= $0.078 \times 11.784 \times 18.1$) billion yuan.

³ Data source: <https://aqli.epic.uchicago.edu/the-index/>.

As for the cost, the total education expenditure in the sample period is 11,870 billion yuan. By subtracting the total cost from the total benefit, we obtain the total net benefit of 4767 billion yuan. The results of the cost-benefit analysis demonstrate that education expenditure significantly improves social welfare, even with the lower bound of benefits.

7. Conclusions and policy implications

Research on the environmental benefit of education expenditure remains limited. This paper explores the effect of education expenditure on air pollution based on county-level panel data in China. Our main conclusions are as follows. (1) Education expenditure negatively affects $PM_{2.5}$ concentration. Our findings are robust after addressing the endogeneity problem, and various robustness tests support our conclusions. (2) The mechanism analysis demonstrates that education expenditure reduces $PM_{2.5}$ concentration through the composition, technique, and income effects. (3) Our empirical results exhibit regional heterogeneity. In contrast to weak regulation areas, the impact of education expenditure is significant in strong regulation areas. Education expenditure plays a greater negative role in key regions than non-key regions. Compared with the eastern and central regions, education expenditure fails to significantly exert its role in the western and northeastern regions. (4) Socioeconomic factors such as human capital, economic development, infrastructure construction, and public service have moderating effects on the role of education expenditure.

Our study sheds new light on the strand of literature regarding the effect of education expenditure by local governments on air pollution. Some policy implications are still worth discussing from fiscal and environmental perspectives. First, our conclusions suggest that education expenditure has a negative effect on the reduction of air pollution, which encourages county governments to emphasize the role of

education expenditure. Due to the long payback cycle, education expenditure may not be adequately guaranteed by local governments. Our findings support that increasing public education investment can simultaneously achieve environmental improvements and economic benefits. Hence, local governments should take the initiative to adjust the structure of fiscal spending and ensure the scale of education expenditure.

Second, the effect of education expenditure on air pollution has regional heterogeneity. Given the regional heterogeneity, policymakers should enact appropriate environmental and education policies considering the different circumstances of each region. In severely polluted areas, environmental regulations and educational finance instruments ought to be simultaneously implemented. Specifically, while ensuring the education budget, county governments need to set standards for air pollutant limits based on local pollution levels. Moreover, fiscal measures such as raising education expenditure might be contemplated in the more economically developed eastern and central regions. Policymakers can improve the education financial security system and fully utilize the role of education expenditure in mitigating air pollution.

Finally, we conclude that socioeconomic factors have moderating effects on the role of education expenditure. On the one hand, improving human capital levels facilitates the role of education expenditure, which encourages the government to increase education investment. Specifically, governments at all levels should raise the proportion of education in fiscal spending to enhance the average level of human capital and strengthen public awareness of environmental protection. At the same time, the government should conduct public education activities on environmental protection to motivate the public to focus on air quality. On the other hand, the development of infrastructure construction and public services is also crucial, suggesting that the government should create a favorable economic and social environment for citizens. Therefore, local governments should optimize the structure of fiscal expenditure and increase public fiscal spending. To this end, the government should simultaneously improve infrastructure construction and

guarantee solid medical and health care to provide the public with convenient living conditions and ample welfare benefits.

The limitation of this paper mainly lies in the level of data. Given the difficulty in accessing county-level data, we could only identify the role of education expenditure but not expenditure on other public goods. In other words, we are incapable of analyzing the relationship between the components of fiscal spending and environmental quality. Moreover, we could not control more factors related to education expenditure at the county level. We endeavor to overcome this difficulty by controlling for county fixed effects and establishing 2SLS models to provide accurate evidence of a causal relationship.

CRediT authorship contribution statement

Nan Zhao: Supervision, Methodology, Conceptualization. **Chenyang Wang:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Chunyan Shi:** Validation, Data curation. **Xiaojie Liu:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A

Table A1

List of cities and provinces labeled as key regions.

Key regions	List of cities and provinces
Jing-Jin-Ji and Surrounding Areas	Beijing, Tianjin, Shijiazhuang, Tangshan, Handan, Xingtai, Baoding, Cangzhou, Langfang, Hengshui, and Xiong'an New Area, Taiyuan, Yangquan, Changzhi, Jincheng, Jinan, Zibo, Jining, Dezhou, Liaocheng, Binzhou, Heze, Zhengzhou, Kaifeng, Anyang, Hebi, Xinxiang, Jiaozuo, Puyang.
The Yangtze River Delta	Shanghai, Jiangsu, Zhejiang and Anhui Provinces.
The Fen and Wei Plain	Jinzhong, Yuncheng, Linfen and Luliang in Shanxi Province, Luoyang and Sanmenxia in Henan Province, Xi'an, Tongchuan, Baoji, Xianyang, Weinan and Yangling demonstration zone in Shaanxi Province.

Note: Xiong'an New Area is a state-level new area under the jurisdiction of Hebei Province. Other regions are provinces or prefecture-level cities.

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